Finite Markov Networks: Matrix Trees, Entropy Production, and Exact Fluctuation Symmetry

HCS-C361 theorem package

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Abstract

For every finite irreducible continuous-time Markov chain with bidirected support, we give a convention-complete proof joining the matrix-tree stationary law, positivity and exact equality criteria for entropy production, a finite-time stationary path-reversal theorem, and transpose symmetry of the entire tilted characteristic polynomial. Total and medium entropy are kept distinct. Rate-function symmetry is stated only under explicit large-deviation and Legendre hypotheses. Exact rational ledgers guard all orientation and sign conventions; they do not replace the analytic proof.

1 Frozen chain and theorem

Let $S = \{0, \ldots, d-1\}$ be finite. For $i \neq j$, assume $q_{ij} > 0$ iff $q_{ji} > 0$, and that the undirected support is connected. There are no recorded self-jumps. On column functions,

$$(Lf)(i) = \sum_{j \neq i} q_{ij}(f(j) - f(i)), \qquad L_{ij} = q_{ij}, \quad L_{ii} = -r_i, \qquad r_i = \sum_{j \neq i} q_{ij}.$$

Let $T \to i$ denote a directed spanning tree whose edges point toward root i, and set $\tau_i = \sum_{T \to i} \prod_{(u \to v) \in T} q_{uv}$.

Theorem 1 *The unique stationary row law is $\pi_i = \tau_i / \sum_k \tau_k$. With*

$$a_{ij} = \pi_i q_{ij}, \quad J_{ij} = a_{ij} - a_{ji}, \quad F_{ij} = \log(a_{ij}/a_{ji}),$$

the entropy-production rate satisfies

$$\sigma = \sum_{i<j} (a_{ij} - a_{ji}) \log \frac{a_{ij}}{a_{ji}} \geq 0.$$

Moreover, $\sigma = 0$, detailed balance, and zero affinity $\sum_{(i,j) \in C} \log(q_{ij}/q_{ji}) = 0$ on every oriented cycle C are equivalent.

Start in π. For a path with state skeleton $i_0, \ldots, i_m$, define

$$\Sigma_T = \log \frac{\pi_{i_0} \prod_{\ell=1}^m q_{i_{\ell-1} i_\ell}}{\pi_{i_m} \prod_{\ell=1}^m q_{i_\ell i_{\ell-1}}}.$$

If Θ reverses time, define the reversed pushforward law $\mathbb{P}_\pi^R = \mathbb{P}_\pi \circ \Theta^{-1} = \mathbb{P}_\pi \circ \Theta$. Then

$$\frac{d\mathbb{P}_\pi}{d\mathbb{P}_\pi^R} = e^{\Sigma_T}, \qquad \mathbb{E} e^{-\Sigma_T} = 1, \qquad \mathbb{E} e^{-\lambda \Sigma_T} = \mathbb{E} e^{-(1-\lambda)\Sigma_T}.$$

More precisely, its law μ_T obeys $\mu_T(B) = \int_{-B} e^{-s} \mu_T(ds)$, where $-B = \{-s : s \in B\}$; a point-probability formula is intended only at atoms. The exponential identity holds for every real λ.

For medium entropy $W_T = \sum \log(q_{ij}/q_{ji})$, the tilted matrix is

$$(L_\lambda)_{ij} = q_{ij}^{1-\lambda} q_{ji}^{\lambda} \quad (i \neq j), \qquad (L_\lambda)_{ii} = -r_i.$$

It satisfies $L_\lambda^\intercal = L_{1-\lambda}$ and hence, for all $z \in \mathbb{C}$,

$$\det(zI - L_\lambda) = \det(zI - L_{1-\lambda}).$$

For every real λ, the SCGF $\psi(\lambda) = \lim_{T \to \infty} T^{-1} \log \mathbb{E}_\pi e^{-\lambda W_T}$ exists, equals the Perron eigenvalue of L_λ, is finite and real analytic, and has the same symmetry. If, additionally, W_T/T obeys a full LDP whose rate equals the Legendre–Fenchel transform $I(a) = \sup_\lambda\{-\lambda a - \psi(\lambda)\}$, then $I(a) - I(-a) = -a$.

2 Proof

For $M = -L^{\mathsf{T}}$, principal-cofactor expansion cancels selections containing directed cycles and leaves precisely the in-arborescence products. Thus $L^{\mathsf{T}}\tau = 0$. Connectivity gives $\tau_i > 0$ for every root; irreducibility makes the nullspace one-dimensional, proving the stationary formula.

The inequality $(x-y)\log(x/y) \geq 0$ for $x, y > 0$ proves nonnegativity edge by edge, with equality exactly when $a_{ij} = a_{ji}$. Detailed balance telescopes around every cycle. Conversely, assume all cycle products are one. Fix o and define along any support path

$$h_i = \prod_{o=i_0\to\cdots\to i_m=i} \frac{q_{i_{\ell-1}i_\ell}}{q_{i_\ell i_{\ell-1}}}.$$

The value is path-independent: concatenate two paths, erase backtracks, and split at repeated vertices into simple cycles. Therefore $h_j/h_i = q_{ij}/q_{ji}$ on every edge. Normalized h satisfies detailed balance and stationarity, so it equals π.

For fixed jump skeleton and holding intervals $s_0, \ldots, s_m$, the path density equals

$$\pi_{i_0} \exp\left(-\sum_{\ell=0}^{m} r_{i_\ell} s_\ell\right) \prod_{\ell=1}^{m} q_{i_{\ell-1}i_\ell}.$$

Reversal only reorders the holding exponential and reverses each jump. Bidirectionality therefore gives the stated Radon–Nikodym derivative against $\mathbb{P}_\pi^R$, including zero-jump paths, and $\Sigma_T \circ \Theta = -\Sigma_T$. Change of variables yields $\mathbb{E}[g(\Sigma_T)] = \mathbb{E}[e^{-\Sigma_T} g(-\Sigma_T)]$; indicators and exponentials give every finite-time identity.

Feynman–Kac multiplication of a jump $i \to j$ by $e^{-\lambda \log(q_{ij}/q_{ji})}$ gives the displayed tilt, whose transpose identity is entrywise. Transposition preserves the full characteristic polynomial. After a scalar shift, Perron–Frobenius supplies a simple real dominant eigenvalue and finite-dimensional Feynman–Kac identifies it with ψ. Since the tilted entries depend analytically on real λ and this eigenvalue is simple, the local branches join to make ψ real analytic on $\mathbb{R}$. Finally, $|\Sigma_T - W_T| \leq 2\max_i |\log \pi_i|$, so total and medium entropy have the same SCGF. Substitution $\mu = 1 - \lambda$ in the explicitly assumed Legendre formula proves the rate symmetry; no rate-function conclusion is drawn without that full-LDP identification.

3 Sharp boundaries and a minimal nonequilibrium cycle

For one state, the empty tree has weight one and all entropy is zero. Every irreducible two-state chain is reversible because stationarity forces $\pi_0 q_{01} = \pi_1 q_{10}$; indeed $\Sigma_T = 0$ pathwise. A nonzero affinity first occurs with three states. Give every clockwise edge rate 2 and every reverse edge rate 1. Then $\pi_i = 1/3$, the clockwise affinity is $3\log 2$, each oriented stationary current is $1/3$, and $\sigma = \log 2$.

Reducibility destroys the unique positive global law and positive rooted-tree weights for every root; the theorem applies after selecting a closed irreducible class. A one-way edge can make the reversed path measure singular and the affinity infinite, so it is not a zero-rate substitution into this theorem. Phantom self-transitions are excluded; diagonals encode holding.